

Polynomial — Grade 6 Ready Type Examples

One Easy, Medium, and Hard sample per Ready Grade 6 type, formatted like printable worksheets
(shared instructions, graphs, and diagrams).

Rectangles with fraction side lengths

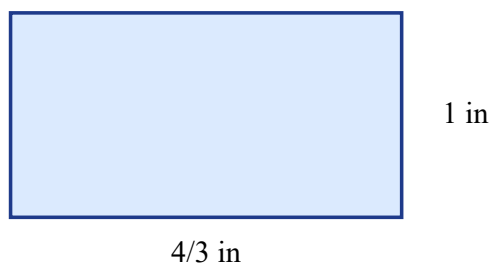
g6_rectangles_with_fraction_side_lengths · Grade 6 — Area and Volume with Fractions ·
gen=g6_fraction_rectangle_area

Name: _____ Date: _____

Find the area.

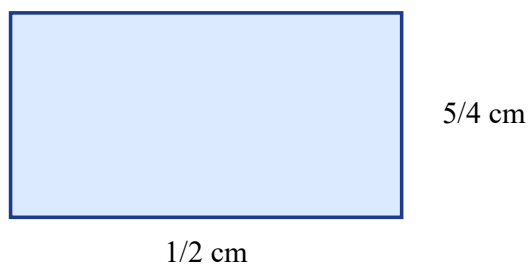
EASY

1. Find the area of the rectangle with base $\frac{4}{3}in$ and height $1in$.



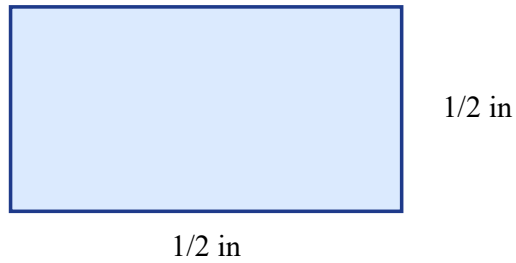
MEDIUM

2. Find the area of the rectangle with base $\frac{1}{2}cm$ and height $\frac{5}{4}cm$.



HARD

3. Find the area of the rectangle with base $\frac{1}{2}in$ and height $\frac{1}{2}in$.



Answer Key

1. **Easy:** $\frac{4}{3}in^2$
2. **Medium:** $\frac{5}{8}cm^2$
3. **Hard:** $\frac{1}{4}in^2$
-

Right rectangular prisms with fraction side lengths

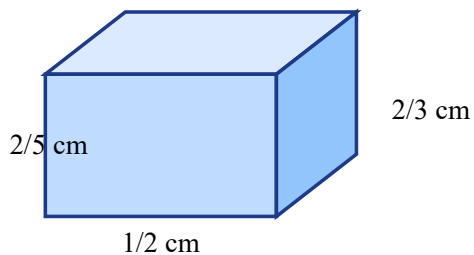
g6_right_rectangular_prisms_with_fraction_side_lengths · Grade 6 – Area and Volume with Fractions · gen=g6_fraction_prism_volume

Name: _____ Date: _____

Find the volume.

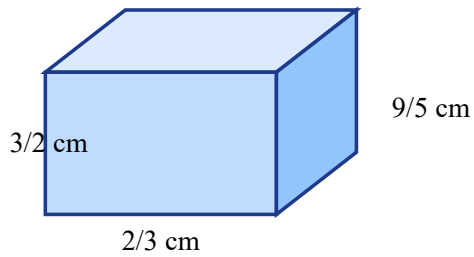
EASY

1. Find the volume of the right rectangular prism with side lengths $\frac{1}{2}cm$, $\frac{2}{3}cm$, $\frac{2}{5}cm$.

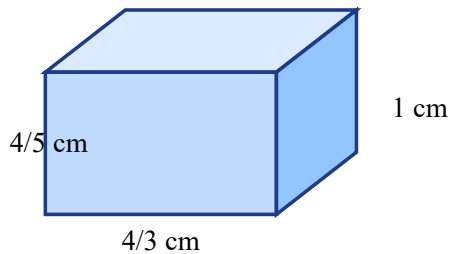


MEDIUM

2. Find the volume of the right rectangular prism with side lengths $\frac{2}{3}cm$, $\frac{9}{5}cm$, $\frac{3}{2}cm$.

**HARD**

3. Find the volume of the right rectangular prism with side lengths $\frac{4}{3}cm$, $1cm$, $\frac{4}{5}cm$.



Answer Key

1. **Easy:** $\frac{2}{15}cm^3$
2. **Medium:** $\frac{9}{5}cm^3$
3. **Hard:** $\frac{16}{15}cm^3$

Triangles with fraction side lengths

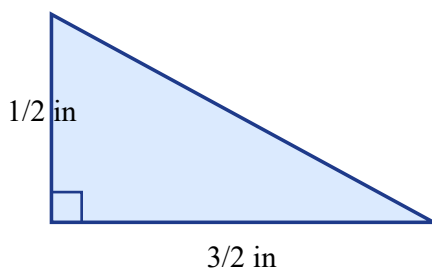
g6_triangles_with_fraction_side_lengths • Grade 6 – Area and Volume with Fractions •
gen=g6_fraction_triangle_area

Name: _____ Date: _____

Find the area.

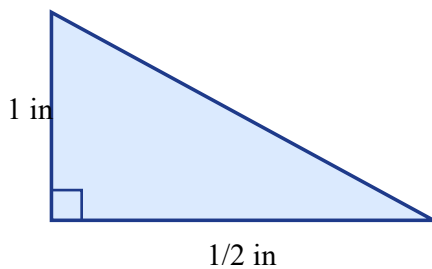
EASY

1. Find the area of the right triangle with base $\frac{3}{2}in$ and height $\frac{1}{2}in$.



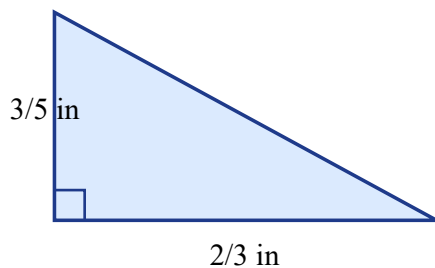
MEDIUM

2. Find the area of the right triangle with base $\frac{1}{2}in$ and height $1in$.



HARD

3. Find the area of the right triangle with base $\frac{2}{3}in$ and height $\frac{3}{5}in$.



Answer Key

1. **Easy:** $\frac{3}{8}in^2$

2. **Medium:** $\frac{1}{4}in^2$

3. **Hard:** $\frac{1}{5}in^2$

Factoring

g6_factoring · Grade 6 – Common Factors and Common Multiples · gen=g6_factoring

Name: _____ Date: _____

Factor.

EASY

1. Write the prime factorization of
21

MEDIUM

2. Write the prime factorization of
14

HARD

3. Write the prime factorization of 105

Answer Key

1. **Easy:** $3 \cdot 7$

2. **Medium:** $2 \cdot 7$

3. **Hard:** $3 \cdot 5 \cdot 7$

GCF and LCM word problems

g6_gcf_and_lcm_word_problems · Grade 6 – Common Factors and Common Multiples ·
gen=g6_gcf_and_lcm_word_problems

Name: _____ Date: _____

Solve the problem.

EASY

1. Event A happens every 20 minutes and Event B every 6 minutes. After how many minutes do they happen together?

MEDIUM

2. Event A happens every 30 minutes and Event B every 30 minutes. After how many minutes do they happen together?

HARD

3. Event A happens every 85 minutes and Event B every 57 minutes. After how many minutes do they happen together?

Answer Key

1. **Easy:** 60
 2. **Medium:** 30
 3. **Hard:** 4845
-

Greatest common factor

g6_greatest_common_factor • Grade 6 – Common Factors and Common Multiples •
gen=g6_greatest_common_factor

Name: _____ Date: _____

Find the GCF.

EASY

1. Find the GCF of 24, 27

MEDIUM

2. Find the GCF of 9, 45, 27

HARD

3. Find the GCF of 49, 7

Answer Key

1. Easy: 3
2. Medium: 9
3. Hard: 7

Least common multiple

g6_least_common_multiple · Grade 6 – Common Factors and Common Multiples ·
gen=g6_least_common_multiple

Name: _____ Date: _____

Find the LCM.

EASY

1. Find the LCM of 24, 12

MEDIUM

2. Find the LCM of 42, 48, 4

HARD

3. Find the LCM of 60, 94

Answer Key

1. Easy: 24
2. Medium: 336
3. Hard: 2820

Distances on the coordinate plane

g6_distances_on_the_coordinate_plane · Grade 6 – Coordinate Plane ·
gen=geo_coordinate_distance

Name: _____ Date: _____

Find the distance.

EASY

1. Find the distance between $(4, -3)$ and $(-4, -3)$.

MEDIUM

2. Find the distance between $(4, 7)$ and $(4, -6)$.

HARD

3. Find the distance between $(4, -9)$ and $(4, -10)$.

Answer Key

1. **Easy:** 8
 2. **Medium:** 13
 3. **Hard:** 1
-

Distances, word problems

g6_coordinate_plane_distances_word_problems · Grade 6 – Coordinate Plane ·
gen=wp_coordinate_distance

Name: _____ Date: _____

Solve the problem.

EASY

1. Point A is at $(-2, 4)$ and point B is at $(5, 7)$ on a coordinate plane. How far apart are the points in units?

MEDIUM

2. Point A is at $(2, 2)$ and point B is at $(10, 9)$ on a coordinate plane. How far apart are the points in units?

HARD

3. Point A is at $(1, 0)$ and point B is at $(9, 8)$ on a coordinate plane. How far apart are the points in units?

Answer Key

1. **Easy:** 8
 2. **Medium:** 11
 3. **Hard:** 11
-

Points on the coordinate plane

g6_points_on_the_coordinate_plane · Grade 6 – Coordinate Plane · gen=plotting_points

Name: _____ Date: _____

Identify the point.

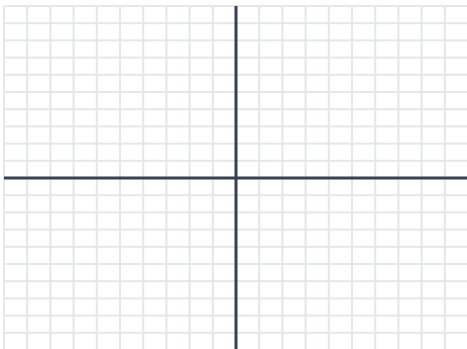
EASY

1. $(0, 3)$



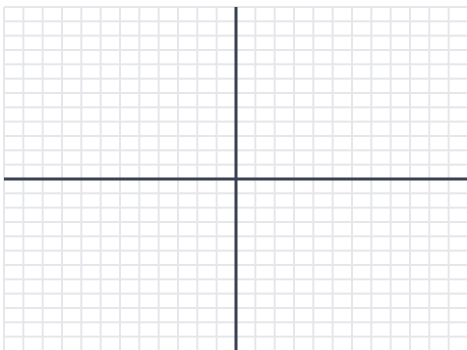
MEDIUM

2. $(-3, 8)$



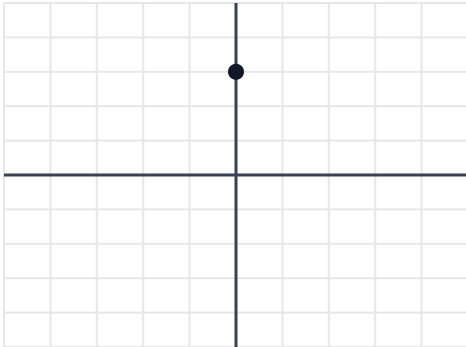
HARD

3. $(6, -4)$

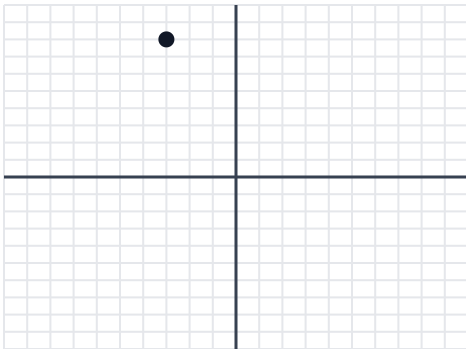


Answer Key

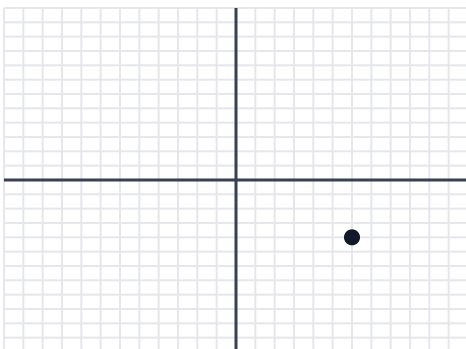
1. **Easy:** $(0, 3)$



2. **Medium:** $(-3, 8)$



3. **Hard:** $(6, -4)$



Shapes and perimeter on the coordinate plane

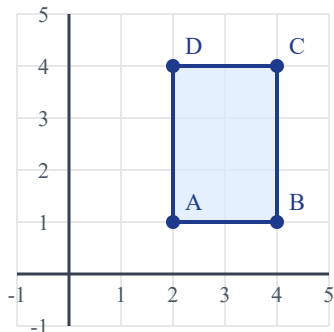
g6_shapes_and_perimeter_on_the_coordinate_plane · Grade 6 – Coordinate Plane ·
gen=g6_coordinate_perimeter

Name: _____ Date: _____

Find the perimeter.

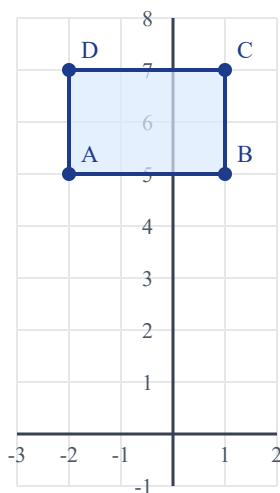
EASY

1. A rectangle on the coordinate plane has vertices $A(2, 1)$, $B(4, 1)$, $C(4, 4)$, $D(2, 4)$. Find the perimeter.



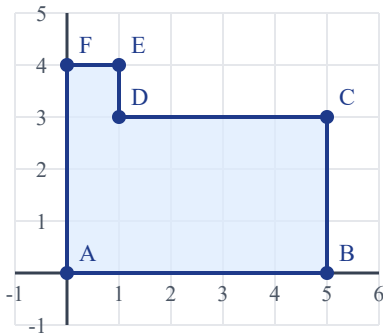
MEDIUM

2. A rectangle on the coordinate plane has vertices $A(-2, 5)$, $B(1, 5)$, $C(1, 7)$, $D(-2, 7)$. Find the perimeter.



HARD

3. A L-shaped polygon on the coordinate plane has vertices $A(0, 0)$, $B(5, 0)$, $C(5, 3)$, $D(1, 3)$, $E(1, 4)$, $F(0, 4)$. Find the perimeter.



Answer Key

1. **Easy:** 10
 2. **Medium:** 10
 3. **Hard:** 18
-

Center and spread

g6_data_center_and_spread • Grade 6 – Data Sets and Distributions • gen=stats_center_spread

Name: _____ Date: _____

Find the measure.

EASY

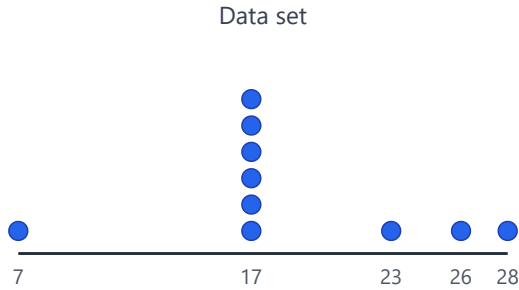
1. Find the mode of the data set: $\{17, 17, 17, 17, 14, 7\}$.

Data set

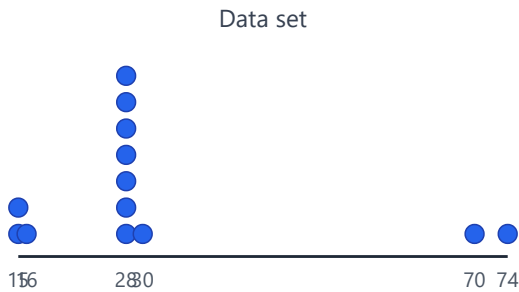


MEDIUM

2. Find the mode of the data set: {17, 17, 17, 17, 17, 17, 26, 28, 23, 7}.

**HARD**

3. Find the mode of the data set: {28, 28, 28, 28, 28, 28, 28, 30, 74, 70, 15, 15, 16}.



Answer Key

1. **Easy:** 17

2. **Medium:** 17

3. **Hard:** 28

Drawing dot plots

g6_drawing_dot_plots · Grade 6 – Data Sets and Distributions · gen=g6_drawing_dot_plot

Name: _____ Date: _____

Draw the dot plot.

EASY

1. Sketch/copy the dot plot for the data set $\{1, 4, 8, 5, 8, 12\}$.

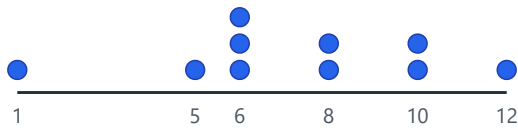
Dot plot to copy



MEDIUM

2. Sketch/copy the dot plot for the data set $\{6, 6, 5, 6, 8, 10, 12, 1, 10, 8\}$.

Dot plot to copy



HARD

3. Sketch/copy the dot plot for the data set $\{8, 10, 5, 1, 4, 2\}$.

Dot plot to copy



Answer Key

1. **Easy:** *(no answer)*
2. **Medium:** *(no answer)*
3. **Hard:** *(no answer)*
-

Drawing histograms

g6_drawing_histograms · Grade 6 – Data Sets and Distributions · gen=g6_drawing_histogram

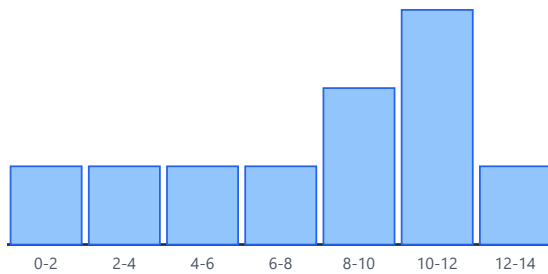
Name: _____ Date: _____

Draw the histogram.

EASY

1. Sketch/copy the histogram for the data set $\{1, 3, 8, 9, 12, 11, 6, 11, 5\}$.

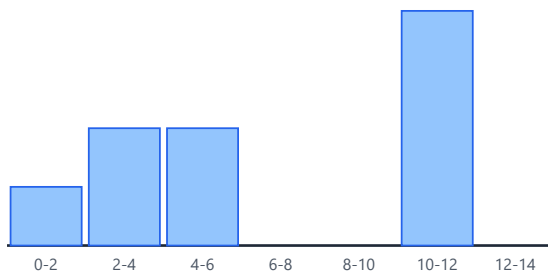
Histogram to copy



MEDIUM

2. Sketch/copy the histogram for the data set $\{10, 1, 10, 2, 10, 3, 4, 5, 11\}$.

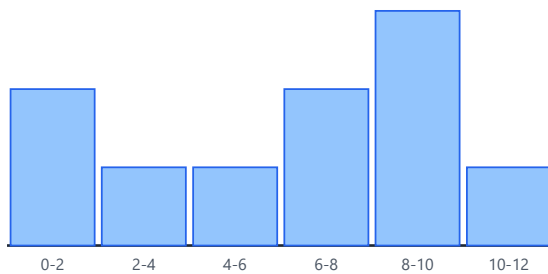
Histogram to copy



HARD

3. Sketch/copy the histogram for the data set $\{8, 10, 2, 1, 7, 9, 1, 7, 4\}$.

Histogram to copy



Answer Key

1. **Easy:** *(no answer)*
2. **Medium:** *(no answer)*
3. **Hard:** *(no answer)*

Interpreting box plots

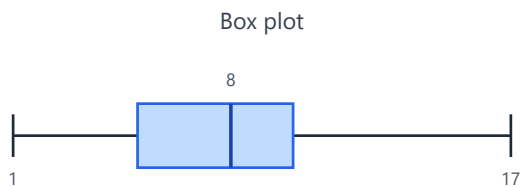
g6_interpreting_box_plots · Grade 6 – Data Sets and Distributions ·
gen=stats_box_plot_basics

Name: _____ Date: _____

Interpret the box plot.

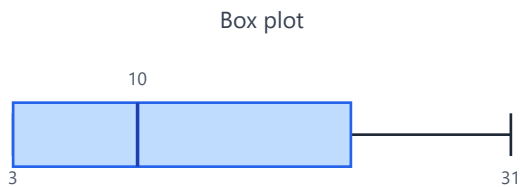
EASY

1. From the box plot, find the interquartile range.



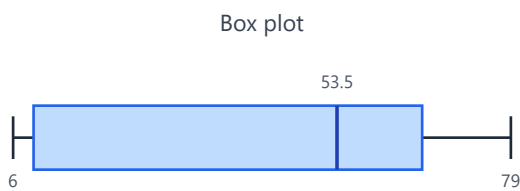
MEDIUM

2. From the box plot, find the median.



HARD

3. From the box plot, find the median.



Answer Key

1. **Easy:** 5
2. **Medium:** 10
3. **Hard:** 53.5

Interpreting dot plots

g6_interpreting_dot_plots · Grade 6 – Data Sets and Distributions · gen=stats_dot_plot_read

Name: _____ Date: _____

Interpret the dot plot.

EASY

1. The dot plot shows the data set below. How many dots are at 6?

Data: {1, 2, 15, 10, 3, 6}

Dot plot



MEDIUM

2. The dot plot shows the data set below. How many dots are at 11?

Data: {36, 13, 18, 9, 15, 20, 11, 40, 20}

Dot plot



HARD

3. The dot plot shows the data set below. How many dots are at 65?

Data: {32, 21, 11, 32, 16, 1, 65, 40}

Dot plot



Answer Key

1. **Easy:** 1

2. **Medium:** 1

3. **Hard:** 1

Interpreting histograms

g6_interpreting_histograms · Grade 6 – Data Sets and Distributions ·
gen=stats_histogram_read

Name: _____ Date: _____

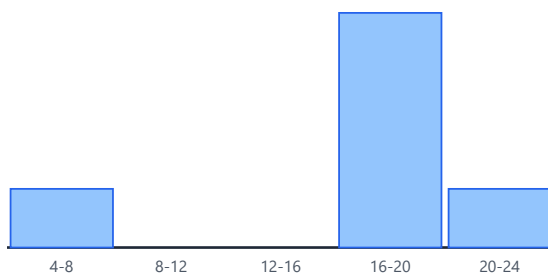
Interpret the histogram.

EASY

1. How many values fall in the interval $[4, 8)$?

Data: {18, 18, 17, 4, 20}

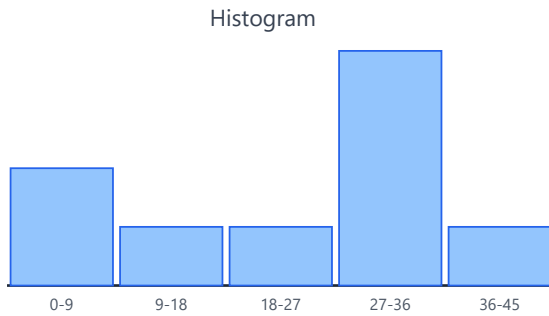
Histogram



MEDIUM

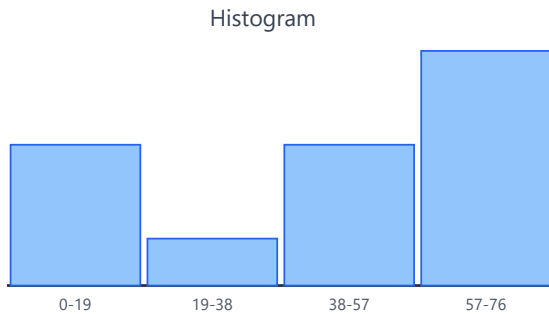
2. How many values fall in the interval $[0, 9)$?

Data: $\{28, 5, 17, 34, 6, 33, 25, 36\}$

**HARD**

3. How many values fall in the interval $[0, 19)$?

Data: $\{24, 58, 44, 70, 45, 15, 70, 12, 58, 72, 56, 6\}$



Answer Key

1. **Easy:** 1

2. **Medium:** 2

3. **Hard:** 3

Decimal addition

g6_decimal_addition · Grade 6 – Decimal Arithmetic · gen=g6_decimal_addition

Name: _____ Date: _____

Add.

EASY

1. $58.9 + 65.4$

MEDIUM

2. $37.04 + 49.83$

HARD

3. $-31.309 + -15.56$

Answer Key

1. **Easy:** 124.3

2. **Medium:** 86.87

3. **Hard:** -46.869

Decimal multiplication

g6_decimal_multiplication · Grade 6 – Decimal Arithmetic · gen=g6_decimal_multiplication

Name: _____ Date: _____

Multiply.

EASY

1. $10 \cdot 0.2$

MEDIUM

2. $1.7 \cdot 3.64$

HARD

3. $0.521 \cdot 2.43$

Answer Key

1. **Easy:** 2

2. **Medium:** 6.188

3. **Hard:** 1.26603

Decimal multiplication with equivalent fractions

g6_decimal_multiplication_with_equivalent_fractions · Grade 6 – Decimal Arithmetic ·
gen=g6_decimal_multiplication

Name: _____ Date: _____

Multiply.

EASY

1. $2 \cdot 0.7$

MEDIUM

2. $1.1 \cdot 6.8$

HARD

3. $0.527 \cdot 8.5$

Answer Key

1. **Easy:** 1.4
 2. **Medium:** 7.48
 3. **Hard:** 4.4795
-

Decimal subtraction

g6_decimal_subtraction · Grade 6 – Decimal Arithmetic · gen=g6_decimal_subtraction

Name: _____ Date: _____

Subtract.

EASY

1. $88.5 - 31.6$

MEDIUM

2. $65.54 - 35.24$

HARD

3. $17.214 - -83.191$

Answer Key

1. **Easy:** 56.9
 2. **Medium:** 30.3
 3. **Hard:** 100.405
-

Dividing decimals by decimals

g6_dividing_decimals_by_decimals · Grade 6 – Decimal Arithmetic · gen=g6_decimal_divide

Name: _____ Date: _____

Divide.

EASY

1. $3.0 \div 0.25$

MEDIUM

2. $2.0 \div 0.2$

HARD

3. $18.0 \div 1.5$

Answer Key

1. **Easy:** 12

2. **Medium:** 10

3. **Hard:** 12

Dividing decimals by whole numbers

g6_dividing_decimals_by_whole_numbers · Grade 6 – Decimal Arithmetic ·
gen=g6_dividing_decimals_by_whole_numbers

Name: _____ Date: _____

Divide.

EASY

1. $28 \div 7$

MEDIUM

2. $31.15 \div 5$

HARD

3. $8.724 \div 3$

Answer Key

1. **Easy:** 4

2. **Medium:** 6.23

3. **Hard:** 2.908

Dividing whole numbers by decimals

g6_dividing_whole_numbers_by_decimals · Grade 6 – Decimal Arithmetic ·
gen=g6_whole_by_decimal_divide

Name: _____ Date: _____

Divide.

EASY

1. $3 \div 0.25$

MEDIUM

2. $11 \div 0.2$

HARD

3. $5 \div 0.2$

Answer Key

1. **Easy:** 12

2. **Medium:** 55

3. **Hard:** 25

Dividing whole numbers that result in decimals

g6_dividing_whole_numbers_that_result_in_decimals · Grade 6 – Decimal Arithmetic ·
gen=g6_dividing_whole_numbers_that_result_in_decimals

Name: _____ Date: _____

Divide.

EASY

1. $48 \div 5$

MEDIUM

2. $167 \div 50$

HARD

3. $5 \div 2$

Answer Key

1. **Easy:** 9.6

2. **Medium:** 3.34

3. **Hard:** 2.5

Long division with remainders

Name: _____ Date: _____

Divide.

EASY

1. $-2 \div 15$

MEDIUM

2. $18 \div -2$

HARD

3. $-37 \div 15$

Answer Key

1. **Easy:** -1

2. **Medium:** -9

3. **Hard:** -3

Dividing fractions

Name: _____ Date: _____

Divide.

EASY

1. $\frac{3}{4} \div \frac{2}{5}$

MEDIUM

2. $\left(\frac{9}{4}\right) / \left(\frac{9}{8}\right)$

HARD

3. $-\frac{22}{5} \div -\frac{1}{2}$

Answer Key

1. **Easy:** $\frac{15}{8}$

2. **Medium:** 2

3. **Hard:** $\frac{44}{5}$

How many groups/times?

g6_how_many_groups_times · Grade 6 – Dividing Fractions · gen=g6_fraction_divide_groups

Name: _____ Date: _____

Solve the problem.

EASY

1. How many groups of $\frac{1}{3}$ are in 2?

MEDIUM

2. How many groups of $\frac{5}{6}$ are in $\frac{5}{3}$?

HARD

3. How many groups of $\frac{7}{8}$ are in $\frac{35}{8}$?

Answer Key

1. **Easy:** 6

2. **Medium:** 2

3. **Hard:** 5

How much in each group/time?

g6_how_much_in_each_group_time · Grade 6 – Dividing Fractions · gen=g6_fraction_divide_each

Name: _____ Date: _____

Solve the problem.

EASY

1. If $\frac{15}{4}$ is shared equally into 5 groups, how much is in each group?

MEDIUM

2. If 2 is shared equally into 6 groups, how much is in each group?

HARD

3. If $\frac{20}{7}$ is shared equally into 2 groups, how much is in each group?

Answer Key

1. **Easy:** $\frac{3}{4}$
 2. **Medium:** $\frac{1}{3}$
 3. **Hard:** $\frac{10}{7}$
-

What fraction of a whole?

g6_what_fraction_of_a_whole · Grade 6 – Dividing Fractions · gen=g6_fraction_of_whole

Name: _____ Date: _____

Solve the problem.

EASY

1. What fraction of a whole is $\frac{1}{2}$ of $\frac{3}{2}$?

MEDIUM

2. What fraction of a whole is $\frac{1}{2}$ of 5?

HARD

3. What fraction of a whole is $\frac{1}{2}$ of $\frac{5}{2}$?

Answer Key

1. **Easy:** $\frac{3}{4}$
 2. **Medium:** $\frac{5}{2}$
 3. **Hard:** $\frac{5}{4}$
-

Equations word problems

g6_equations_word_problems · Grade 6 – Equations · gen=wp_one_step_equation

Name: _____ Date: _____

Solve the problem.

EASY

1. Wade collected 5 equal donations totaling \$15. How much was each donation?

MEDIUM

2. Tessa had some money, spent \$8, and has \$41 left. How much did Tessa start with?

HARD

3. Brian had some money, received \$12, and now has \$29. How much did Brian start with?

Answer Key

1. Easy: \$3
 2. Medium: \$49
 3. Hard: \$17
-

Hanger diagrams

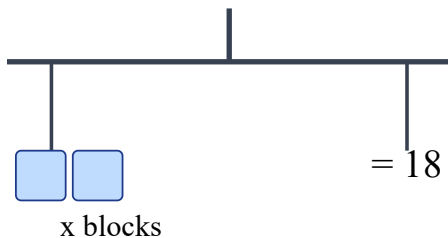
g6_equations_hanger_diagrams · Grade 6 – Equations · gen=g6_equations_hanger_diagrams

Name: _____ Date: _____

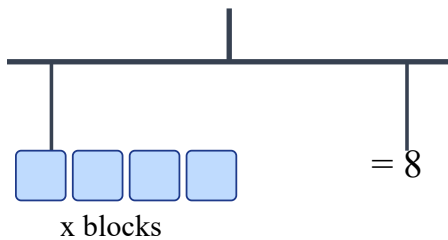
Solve the problem.

EASY

1. Use the diagram to solve $2x = 18$.

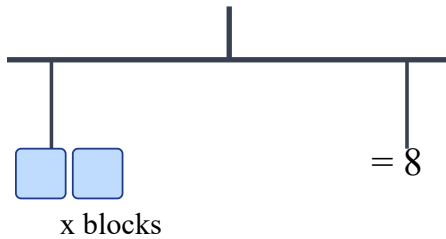
**MEDIUM**

2. Use the diagram to solve $4x = 8$.



HARD

3. Use the diagram to solve $2x = 8$.



Answer Key

1. **Easy:** 9

2. **Medium:** 2

3. **Hard:** 4

Solutions to equations

g6_solutions_to_equations · Grade 6 – Equations · gen=check_equation_solution

Name: _____ Date: _____

Determine if the value is a solution.

EASY

1. Is $x = 5$ a solution of $8x + 5 = 21$?

MEDIUM

2. Is $x = 3$ a solution of $7x + 16 = 51$?

HARD

3. Is $x = 9$ a solution of $8x + 17 = 97$?

Answer Key

1. **Easy:** no

2. **Medium:** no

3. **Hard:** no

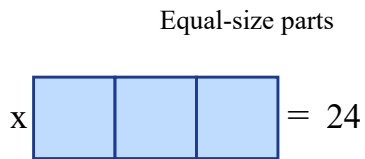
Tape diagrams

Name: _____ Date: _____

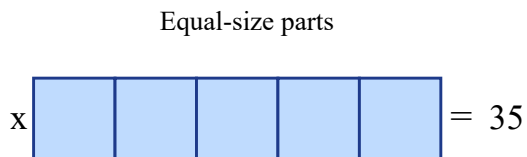
Solve the problem.

EASY

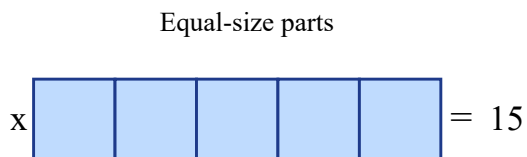
1. Use the diagram to solve $3x = 24$.

**MEDIUM**

2. Use the diagram to solve $5x = 35$.

**HARD**

3. Use the diagram to solve $5x = 15$.



Answer Key

1. **Easy:** 8
2. **Medium:** 7
3. **Hard:** 3

Constant rate equations

g6_constant_rate_equations · Grade 6 – Equations as Relationships between Two Variables ·
gen=write_one_step_equation

Name: _____ Date: _____

Write the equation.

EASY

1. A bike travels at 2 miles per hour. Write an equation for the distance d after 5 hours, then find d .

MEDIUM

2. A bike travels at 6 miles per hour. Write an equation for the distance d after 5 hours, then find d .

HARD

3. A bike travels at 12 miles per hour. Write an equation for the distance d after 4 hours, then find d .

Answer Key

1. **Easy:** $d = 2 \cdot 5$; $d = 10$
2. **Medium:** $d = 6 \cdot 5$; $d = 30$
3. **Hard:** $d = 12 \cdot 4$; $d = 48$

Equations for other relationships

g6_equations_for_other_relationships · Grade 6 – Equations as Relationships between Two Variables · gen=write_one_step_equation

Name: _____ Date: _____

Write the equation.

EASY

1. A bike travels at 5 miles per hour. Write an equation for the distance d after 3 hours, then find d .

MEDIUM

2. A bike travels at 8 miles per hour. Write an equation for the distance d after 4 hours, then find d .

HARD

3. A bike travels at 10 miles per hour. Write an equation for the distance d after 6 hours, then find d .

Answer Key

1. Easy: $d = 5 \cdot 3$; $d = 15$
2. Medium: $d = 8 \cdot 4$; $d = 32$
3. Hard: $d = 10 \cdot 6$; $d = 60$
-

Equivalent ratio equations

g6_equivalent_ratio_equations · Grade 6 – Equations as Relationships between Two Variables ·
gen=solving_proportions

Name: _____ Date: _____

Write the equation.

EASY

1. $\frac{4}{8} = \frac{x}{12}$

MEDIUM

2. $\frac{5}{2} = \frac{x}{4}$

HARD

3. $\frac{x}{5} = \frac{35}{2}$

Answer Key

1. **Easy:** 6
2. **Medium:** 10
3. **Hard:** 14

Combining like terms

g6_combining_like_terms · Grade 6 – Equivalent Expressions · gen=g6_combining_like_terms

Name: _____ Date: _____

Simplify.

EASY

1. $1x + 3x + 1x$

MEDIUM

2. $1x + 6x + 1x$

HARD

3. $22x + 5x + 3x + 9x + 9x$

Answer Key

1. **Easy:** $5x$
2. **Medium:** $8x$
3. **Hard:** $48x$

Distributive property with area diagrams, algebraic

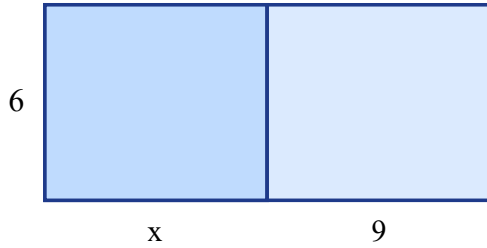
g6_distributive_property_area_diagrams_algebraic · Grade 6 – Equivalent Expressions ·
gen=g6_area_model_algebraic

Name: _____ Date: _____

Simplify.

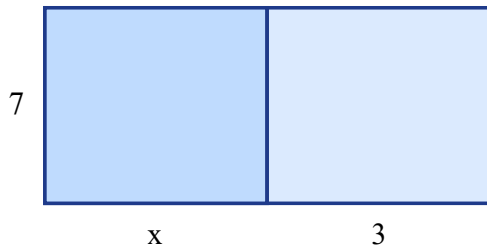
EASY

1. Use the area model to expand $6(x + 9)$.



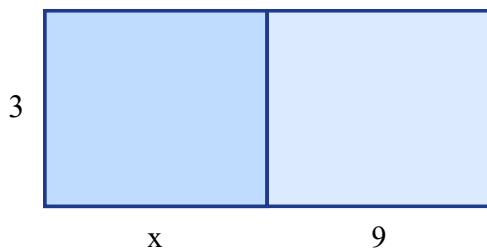
MEDIUM

2. Use the area model to expand $7(x + 3)$.



HARD

3. Use the area model to expand $3(x + 9)$.



Answer Key

1. **Easy:** $6x + 54$

2. **Medium:** $7x + 21$

3. **Hard:** $3x + 27$

Distributive property, algebraic

g6_distributive_property_algebraic · Grade 6 – Equivalent Expressions ·
gen=distributive_property_algebraic

Name: _____ Date: _____

Simplify.

EASY

1. $4(x - 4)$

MEDIUM

2. $2(x + 6)$

HARD

3. $-4(x + 6)$

Answer Key

1. **Easy:** $4x - 16$

2. **Medium:** $2x + 12$

3. **Hard:** $-4x - 24$

Hanger diagrams

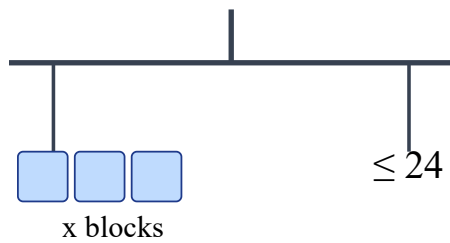
g6_inequalities_hanger_diagrams · Grade 6 – Inequalities ·
gen=g6_inequalities_hanger_diagrams

Name: _____ Date: _____

Solve the problem.

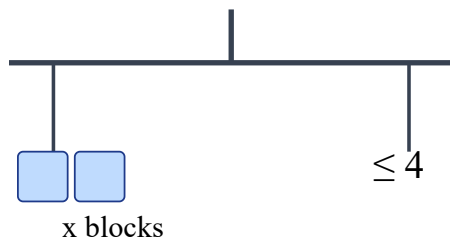
EASY

1. Use the diagram to solve $3x \leq 24$.



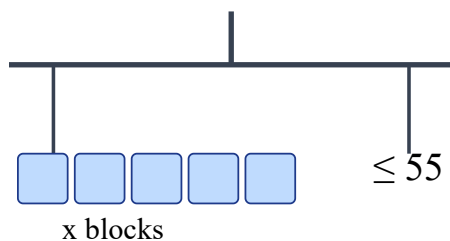
MEDIUM

2. Use the diagram to solve $2x \leq 4$.



HARD

3. Use the diagram to solve $5x \leq 55$.



Answer Key

1. Easy: $x \leq 8$

2. Medium: $x \leq 2$

3. Hard: $x \leq 11$

Name: _____ Date: _____

Solve the problem.

EASY

1. Felix has scored a total of 131 points on 5 tests and wants at least 146 points after one more test. What is the minimum score needed on the next test?

MEDIUM

2. Lucia has scored a total of 116 points on 5 tests and wants at least 134 points after one more test. What is the minimum score needed on the next test?

HARD

3. Will has scored a total of 120 points on 4 tests and wants at least 160 points after one more test. What is the minimum score needed on the next test?

Answer Key

1. Easy: 15
2. Medium: 18
3. Hard: 40

Solutions to inequalities

Name: _____ Date: _____

Graph the following on the number line.

EASY

1. $x \geq -5$



MEDIUM

2. $x < -10$



HARD

3. $x > 5$



Answer Key

1. Easy: $x \geq -5$



2. Medium: $x < -10$



3. Hard: $x > 5$



Solving and graphing one-step inequalities

g6_solving_and_graphing_one_step_inequalities · Grade 6 – Inequalities ·
gen=one_step_inequalities

Name: _____ Date: _____

Solve and graph on the number line.

EASY

1. $\frac{x}{3} > 3$



MEDIUM

2. $x + 5 \geq 3$



HARD

3. $\frac{x}{19} \leq -\frac{13}{57}$

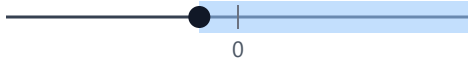


Answer Key

1. Easy: $x > 9$



2. Medium: $x \geq -2$



3. Hard: $x \leq -\frac{13}{3}$



Writing and graphing inequalities

g6_writing_and_graphing_inequalities · Grade 6 – Inequalities ·
gen=graph_single_variable_inequality

Name: _____ Date: _____

Write and graph on the number line.

EASY

1. $x > 7$



MEDIUM

2. $x \geq -8$



HARD

3. $x > 16$

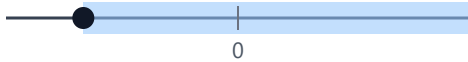


Answer Key

1. **Easy:** $x > 7$



2. **Medium:** $x \geq -8$



3. **Hard:** $x > 16$



Absolute values

g6_absolute_values · Grade 6 – Negative Numbers and Absolute Value · gen=g6_absolute_values

Name: _____ Date: _____

Find the absolute value.

EASY

1. $|-3|$

MEDIUM

2. $|3|$

HARD

3. $|20|$

Answer Key

1. **Easy:** 3

2. **Medium:** 3

3. **Hard:** 20

Comparing numbers

g6_comparing_numbers · Grade 6 – Negative Numbers and Absolute Value ·
gen=g6_comparing_numbers

Name: _____ Date: _____

Compare the numbers.

EASY

1. $7.6 ? -9$

MEDIUM

2. $\frac{1}{3} ? -16$

HARD

3. $-\frac{31}{2} ? -9$

Answer Key

1. **Easy:** >
2. **Medium:** >
3. **Hard:** <

Comparing with absolute values

g6_comparing_with_absolute_values · Grade 6 – Negative Numbers and Absolute Value ·
gen=g6_comparing_with_absolute_values

Name: _____ Date: _____

Compare the values.

EASY

1. $|-10|$? $|-8|$

MEDIUM

2. $|-9|$? $|-8|$

HARD

3. $|-27|$? $|50|$

Answer Key

1. **Easy:** >
2. **Medium:** >
3. **Hard:** <

Number line word problems

g6_number_line_word_problems · Grade 6 – Negative Numbers and Absolute Value ·
gen=wp_number_line

Name: _____ Date: _____

Solve the problem.

EASY

1. The temperature was -10°F and rose 8°F . What is the new temperature?

MEDIUM

2. The temperature was -13°F and dropped 4°F . What is the new temperature?

HARD

3. The temperature was 9°F and dropped 4°F . What is the new temperature?
-

Answer Key

1. **Easy:** -2
2. **Medium:** -17
3. **Hard:** 5
-

Numbers on a number line

g6_numbers_on_a_number_line · Grade 6 – Negative Numbers and Absolute Value ·
gen=number_line_plot

Name: _____ Date: _____

Plot the following numbers on the number line.

EASY

1. -6

**MEDIUM**

2. -2



HARD

3. -7



Answer Key

1. **Easy:** -6



2. **Medium:** -2



3. **Hard:** -7



Opposites of numbers

g6_opposites_of_numbers • Grade 6 – Negative Numbers and Absolute Value •
gen=g6_opposites_of_numbers

Name: _____ Date: _____

Find the opposite.

EASY

1. Find the opposite of -9

MEDIUM

2. Find the opposite of -18

HARD

3. Find the opposite of 49

Answer Key

1. **Easy:** 9

2. **Medium:** 18

3. **Hard:** -49

Ordering numbers

g6_ordering_numbers · Grade 6 – Negative Numbers and Absolute Value ·
gen=g6_ordering_numbers

Name: _____ Date: _____

Order the numbers.

EASY

1. Order from least to greatest: $\frac{1}{3}$,
 -2 , $\frac{9}{11}$

MEDIUM

2. Order from least to greatest: $\frac{11}{12}$,
 -16 , 0.46

HARD

3. Order from least to greatest: $\frac{5}{2}$, $-\frac{8}{11}$, 29

Answer Key

1. **Easy:** -2 , $\frac{1}{3}$, $\frac{9}{11}$

2. **Medium:** -16 , 0.46 , $\frac{11}{12}$

3. **Hard:** $-\frac{8}{11}$, $\frac{5}{2}$, 29

Ordering with absolute values

g6_ordering_with_absolute_values · Grade 6 – Negative Numbers and Absolute Value ·
gen=g6_ordering_with_absolute_values

Name: _____ Date: _____

Order the values.

EASY

1. Order from least to greatest
absolute value: 4, -9 , 0

MEDIUM

2. Order from least to greatest
absolute value: 17, -16 , -5

HARD

3. Order from least to greatest absolute value: -41 , -3 , -42

Answer Key

1. **Easy:** 0, 4, -9
2. **Medium:** -5 , -16 , 17
3. **Hard:** -3 , -41 , -42

Distributive property, numeric

g6_distributive_property_numeric · Grade 6 – Numeric Expressions, Exponents, and the Order of Operations · gen=distributive_property

Name: _____ Date: _____

Evaluate.

EASY

1. $3(1 - 7)$

MEDIUM

2. $8(1 - 1)$

HARD

3. $8(1 - 3)$

Answer Key

1. **Easy:** -18
2. **Medium:** 0
3. **Hard:** -16

Name: _____ Date: _____

Evaluate.

EASY

1. $4 + 5 \cdot 5 - 4$

MEDIUM

2. $2 + 8 \cdot 5 - 6$

HARD

3. $9 + 4 \cdot 2 - 2$

Answer Key

1. **Easy:** 25

2. **Medium:** 36

3. **Hard:** 15

Numeric expressions with exponents

Name: _____ Date: _____

Evaluate.

EASY

1. $3^2 + 3$

MEDIUM

2. $2^3 + 3$

HARD

3. $5^3 + 10$

Answer Key

1. **Easy:** 12

2. **Medium:** 11

3. **Hard:** 135

Properties of addition and multiplication

g6_properties_of_addition_and_multiplication · Grade 6 – Numeric Expressions, Exponents, and the Order of Operations · gen=g6_integer_multiply

Name: _____ Date: _____

Identify the property.

EASY

1. $-6 \cdot 10$

MEDIUM

2. $12 \cdot 17$

HARD

3. $1 \cdot 28$

Answer Key

1. **Easy:** -60

2. **Medium:** 204

3. **Hard:** 28

Writing numeric expressions

g6_writing_numeric_expressions · Grade 6 – Numeric Expressions, Exponents, and the Order of Operations · gen=writing_numeric_expressions

Name: _____ Date: _____

Write the expression.

EASY

1. Write an expression for 4 times the sum of 10 and 3.

MEDIUM

2. Write an expression for the product of 12 and 8.

HARD

3. Write an expression for 7 times the sum of 9 and 3.

Answer Key

1. **Easy:** $4(10 + 3)$
2. **Medium:** $12 \cdot 8$
3. **Hard:** $7(9 + 3)$

Finding percents with equivalent fractions

g6_finding_percents_with_equivalent_fractions · Grade 6 – Percents · gen=percents

Name: _____ Date: _____

Find the percent.

EASY

1. 16 is what percent of 57?

MEDIUM

2. 1 is 6% of what number?

HARD

3. 54.2 is what percent of 83?

Answer Key

1. **Easy:** 29%
2. **Medium:** 16.67
3. **Hard:** 65.3%

Introduction to percents

g6_introduction_to_percents · Grade 6 – Percents · gen=percents

Name: _____ Date: _____

Find the percent.

EASY

1. What is 28% of 93?

MEDIUM

2. 23 is what percent of 96?

HARD

3. What is 75.2% of 436?

Answer Key

1. **Easy:** 26
 2. **Medium:** 24%
 3. **Hard:** 327.9
-

Relating percents, fractions, and decimals

g6_relating_percents_fractions_and_decimals · Grade 6 – Percents ·
gen=g6_relating_percents_fractions_and_decimals

Name: _____ Date: _____

Write the equivalent value.

EASY

1. Write $\frac{1}{2}$ as a decimal.

MEDIUM

2. Write $\frac{3}{100}$ as a decimal.

HARD

3. Write $\frac{9}{20}$ as a decimal.

Answer Key

1. **Easy:** 0.5
 2. **Medium:** 0.03
 3. **Hard:** 0.45
-

Solving percent problems with formulas

g6_solving_percent_problems_with_formulas · Grade 6 – Percents · gen=percents

Name: _____ Date: _____

Solve the problem.

EASY

1. What is 23% of 25?

MEDIUM

2. 53 is 19% of what number?

HARD

3. What is 15.5% of 172?

Answer Key

1. **Easy:** 6

2. **Medium:** 278.9

3. **Hard:** 26.66

Kites

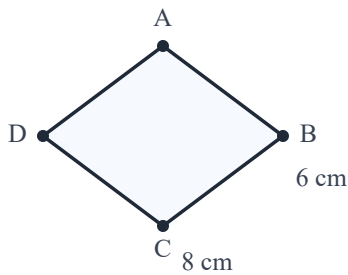
g6_kites · Grade 6 – Polygons · gen=geo_kite_area

Name: _____ Date: _____

Find the area.

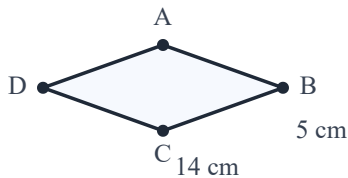
EASY

1. Find the area of the kite.



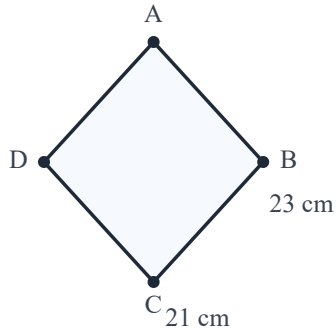
MEDIUM

2. Find the area of the kite.



HARD

3. Find the area of the kite.



Answer Key

1. **Easy:** 24 cm^2
 2. **Medium:** 35 cm^2
 3. **Hard:** 241.5 cm^2
-

Parallelograms

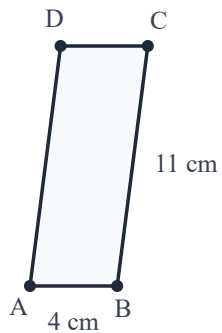
g6_parallelograms · Grade 6 – Polygons · gen=geo_parallelogram_area

Name: _____ Date: _____

Find the area.

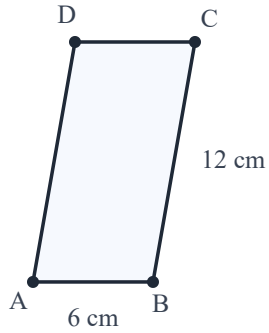
EASY

1. Find the area of the parallelogram.

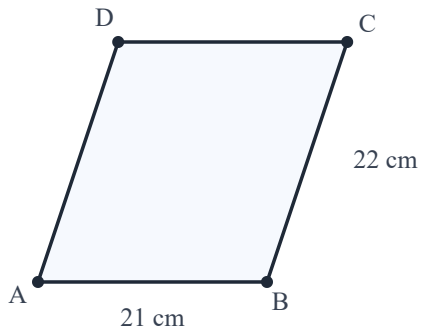


MEDIUM

2. Find the area of the parallelogram.

**HARD**

3. Find the area of the parallelogram.



Answer Key

1. Easy: 44 cm^2
2. Medium: 72 cm^2
3. Hard: 462 cm^2

Parallelograms, understanding area formula

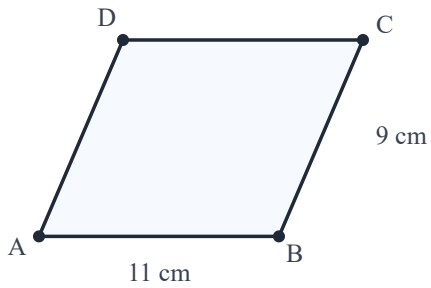
g6_parallelograms_understanding_area_formula • Grade 6 – Polygons •
gen=geo_parallelogram_area

Name: _____ Date: _____

Find the area.

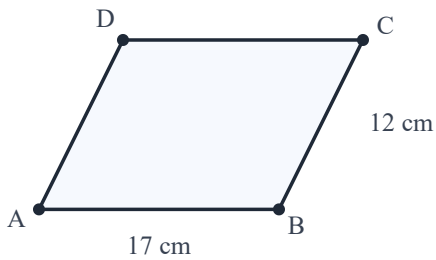
EASY

1. Find the area of the parallelogram.



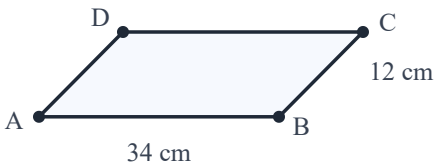
MEDIUM

2. Find the area of the parallelogram.



HARD

3. Find the area of the parallelogram.



Answer Key

1. **Easy:** 99 cm^2

2. **Medium:** 204 cm^2

3. **Hard:** 408 cm^2

Polygons and shaded regions

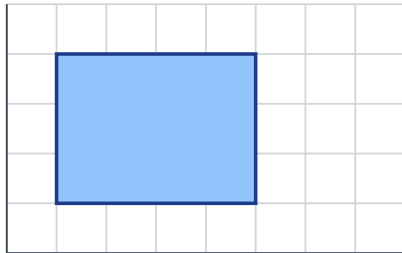
g6_polygons_and_shaded_regions • Grade 6 – Polygons • gen=g6_shaded_polygon_area

Name: _____ Date: _____

Find the area.

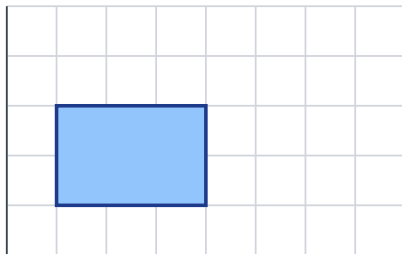
EASY

1. Find the area of the shaded region on the grid.



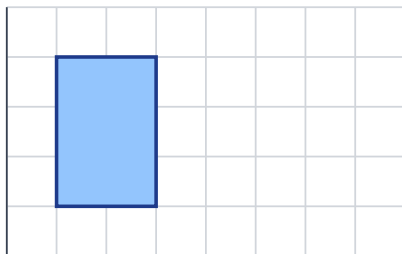
MEDIUM

2. Find the area of the shaded region on the grid.



HARD

3. Find the area of the shaded region on the grid.



Answer Key

1. **Easy:** 12 square units
2. **Medium:** 6 square units
3. **Hard:** 6 square units

Polygons on a grid or coordinate plane

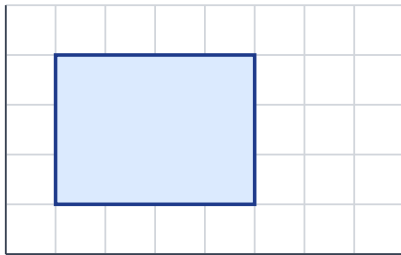
g6_polygons_on_a_grid_or_coordinate_plane · Grade 6 – Polygons · gen=g6_polygon_grid_area

Name: _____ Date: _____

Find the area.

EASY

1. Find the area of the polygon on the grid.



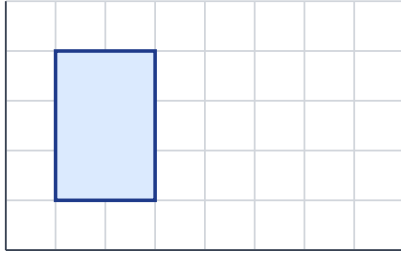
MEDIUM

2. Find the area of the polygon on the grid.



HARD

3. Find the area of the polygon on the grid.



Answer Key

1. **Easy:** 12 square units
 2. **Medium:** 12 square units
 3. **Hard:** 6 square units
-

Trapezoids

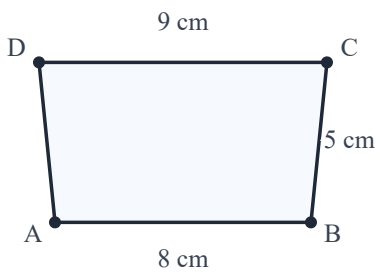
g6_trapezoids · Grade 6 – Polygons · gen=geo_trapezoid_area

Name: _____ Date: _____

Find the area.

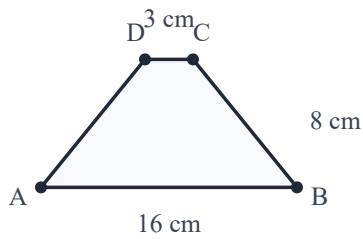
EASY

1. Find the area of the trapezoid.

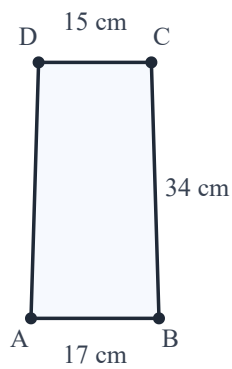


MEDIUM

2. Find the area of the trapezoid.

**HARD**

3. Find the area of the trapezoid.



Answer Key

1. **Easy:** 42.5 cm^2

2. **Medium:** 76 cm^2

3. **Hard:** 544 cm^2

Triangles

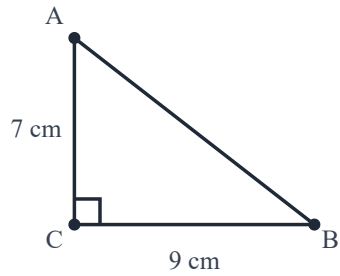
g6_triangles · Grade 6 – Polygons · gen=geo_triangle_area

Name: _____ Date: _____

Find the area.

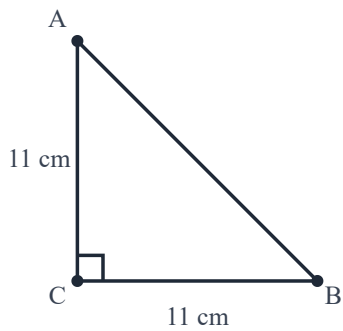
EASY

1. Find the area of $\triangle ABC$.



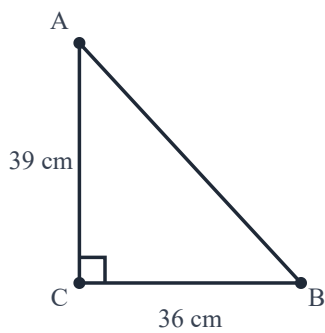
MEDIUM

2. Find the area of $\triangle ABC$.



HARD

3. Find the area of $\triangle ABC$.



Answer Key

1. **Easy:** 31.5 cm^2

2. **Medium:** 60.5 cm^2

3. **Hard:** 702 cm^2

Triangles, understanding area formula

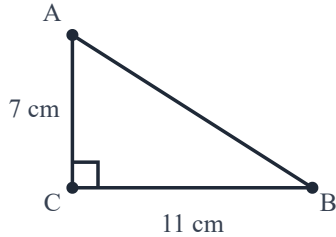
g6_triangles_understanding_area_formula · Grade 6 – Polygons · gen=geo_triangle_area

Name: _____ Date: _____

Find the area.

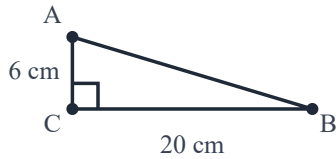
EASY

1. Find the area of $\triangle ABC$.



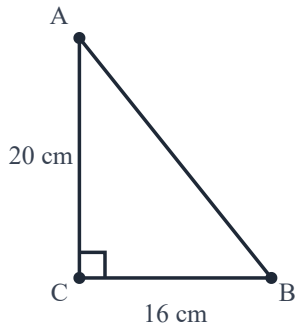
MEDIUM

2. Find the area of $\triangle ABC$.



HARD

3. Find the area of $\triangle ABC$.



Answer Key

1. Easy: 38.5 cm^2

2. Medium: 60 cm^2

3. Hard: 160 cm^2

Classifying and naming

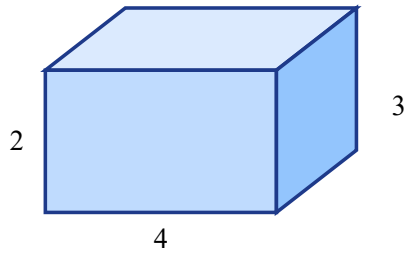
g6_classifying_and_naming · Grade 6 – Polyhedra · gen=g6_classify_polyhedron

Name: _____ Date: _____

Classify the polyhedron.

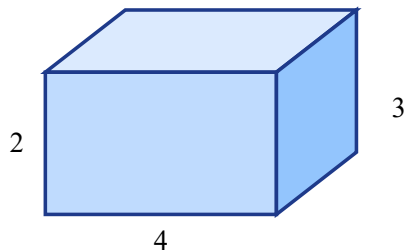
EASY

1. Classify the polyhedron shown.



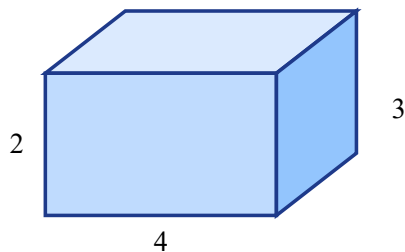
MEDIUM

2. Classify the polyhedron shown.



HARD

3. Classify the polyhedron shown.



Answer Key

1. **Easy:** rectangular prism
 2. **Medium:** rectangular prism
 3. **Hard:** rectangular prism
-

Formulas for volume and surface area of a cube

g6_formulas_for_volume_and_surface_area_of_a_cube · Grade 6 – Polyhedra ·
gen=geo_solid_volume_surface

Name: _____ Date: _____

Find the volume or surface area.

EASY

1. Find the surface area of a $5 \times 9 \times 5$ cm rectangular prism.

MEDIUM

2. Find the volume of a cylinder with radius 7 cm and height 3 cm.

HARD

3. Find the volume of a cylinder with radius 5 cm and height 36 cm.
-

Answer Key

1. **Easy:** 230 cm^2
 2. **Medium:** $49\pi \cdot 3 \text{ cm}^3$
 3. **Hard:** $25\pi \cdot 36 \text{ cm}^3$
-

Identifying invalid nets

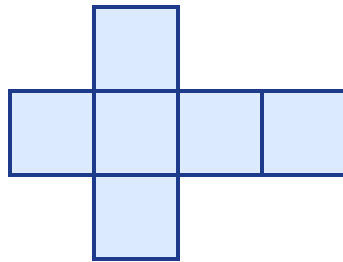
g6_identifying_invalid_nets · Grade 6 – Polyhedra · gen=g6_invalid_cube_net

Name: _____ Date: _____

Determine if the net is valid.

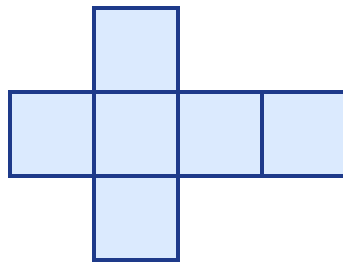
EASY

1. Does this arrangement form a valid net for a cube?



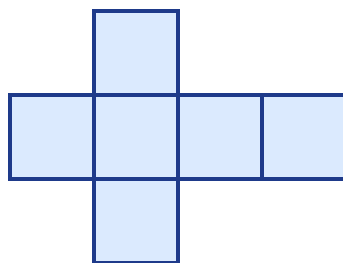
MEDIUM

2. Does this arrangement form a valid net for a cube?



HARD

3. Does this arrangement form a valid net for a cube?



Answer Key

1. **Easy:** No
 2. **Medium:** No
 3. **Hard:** No
-

Isometric sketching

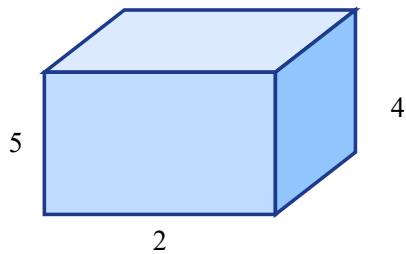
g6_isometric_sketching · Grade 6 – Polyhedra · gen=g6_isometric_solid

Name: _____ Date: _____

Sketch the solid.

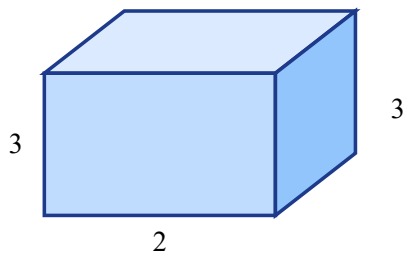
EASY

1. Sketch/copy the rectangular prism shown in the isometric drawing.



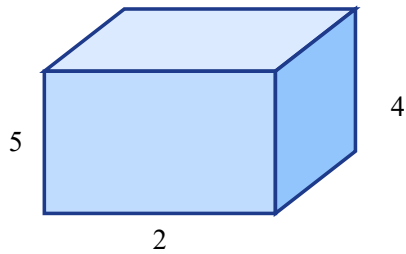
MEDIUM

2. Sketch/copy the rectangular prism shown in the isometric drawing.



HARD

3. Sketch/copy the rectangular prism shown in the isometric drawing.



Answer Key

1. **Easy:** *(no answer)*
2. **Medium:** *(no answer)*
3. **Hard:** *(no answer)*

Nets

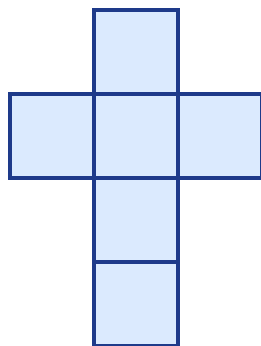
g6_nets • Grade 6 – Polyhedra • gen=g6_cube_nets

Name: _____ Date: _____

Identify the net.

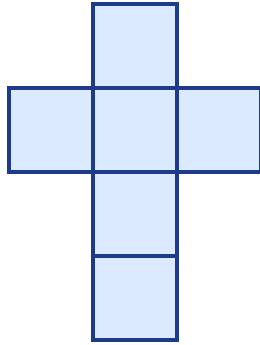
EASY

1. This net folds to make what solid?



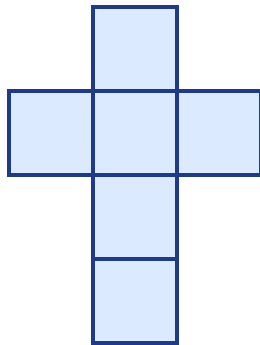
MEDIUM

2. This net folds to make what solid?



HARD

3. This net folds to make what solid?



Answer Key

1. **Easy:** cube
2. **Medium:** cube
3. **Hard:** cube

Nets and surface area

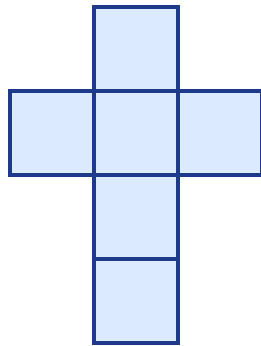
g6_nets_and_surface_area · Grade 6 – Polyhedra · gen=g6_cube_net_surface_area

Name: _____ Date: _____

Find the surface area.

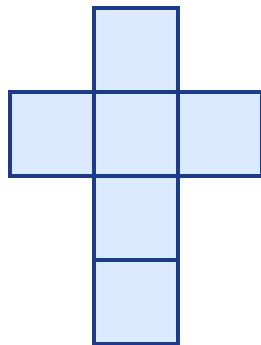
EASY

1. The net is for a cube with edge length 5 units. Find its surface area.



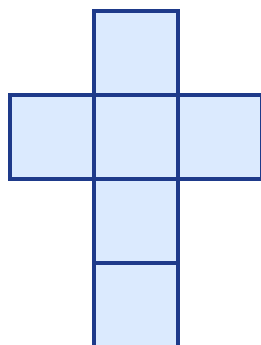
MEDIUM

2. The net is for a cube with edge length 2 units. Find its surface area.



HARD

3. The net is for a cube with edge length 8 units. Find its surface area.



Answer Key

1. **Easy:** 150 square units
2. **Medium:** 24 square units
3. **Hard:** 384 square units

Nets and surface area on a grid

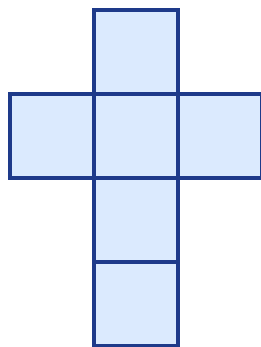
g6_nets_and_surface_area_on_a_grid · Grade 6 – Polyhedra · gen=g6_cube_net_grid_surface_area

Name: _____ Date: _____

Find the surface area.

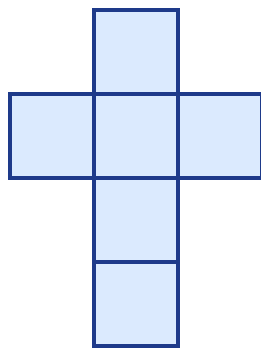
EASY

1. The net is for a cube with edge length 4 units. Find its surface area.



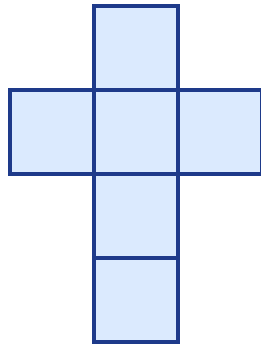
MEDIUM

2. The net is for a cube with edge length 5 units. Find its surface area.



HARD

3. The net is for a cube with edge length 3 units. Find its surface area.



Answer Key

1. **Easy:** 96 square units
 2. **Medium:** 150 square units
 3. **Hard:** 54 square units
-

Volume and surface area using isometric drawings

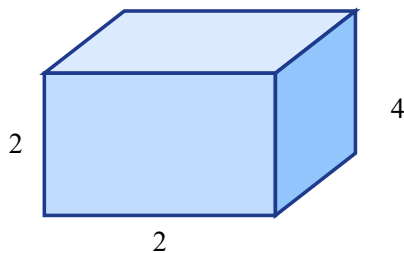
g6_volume_and_surface_area_using_isometric_drawings · Grade 6 – Polyhedra ·
gen=g6_isometric_measure

Name: _____ Date: _____

Find the volume or surface area.

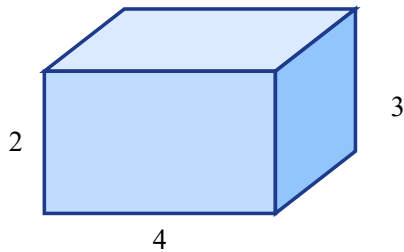
EASY

1. Use the drawing to find the volume.

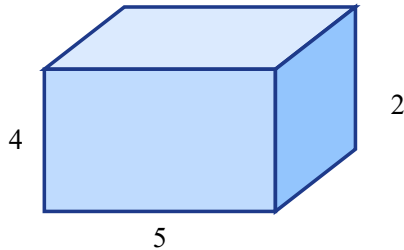


MEDIUM

2. Use the drawing to find the volume.

**HARD**

3. Use the drawing to find the volume.



Answer Key

1. **Easy:** 16 cubic units
2. **Medium:** 24 cubic units
3. **Hard:** 40 cubic units

Comparing rates

g6_comparing_rates • Grade 6 – Rates • gen=g6_unit_rates

Name: _____ Date: _____

Compare the rates.

EASY

1. \$14 for 2 pounds of fruit. Find the unit price in dollars per pound.

MEDIUM

2. A car travels 10 miles in 5 hours. Find the unit rate in miles per hour.

HARD

3. A student reads 40 pages in 5 minutes. Find the unit rate in pages per minute.
-

Answer Key

1. **Easy:** \$7
2. **Medium:** 2
3. **Hard:** 8
-

Converting units

g6_converting_units · Grade 6 – Rates · gen=g6_unit_rates

Name: _____ Date: _____

Convert the units.

EASY

1. \$9 for 3 pounds of fruit. Find the unit price in dollars per pound.

MEDIUM

2. A student reads 8 pages in 4 minutes. Find the unit rate in pages per minute.

HARD

3. A car travels 100 miles in 10 hours. Find the unit rate in miles per hour.
-

Answer Key

1. **Easy:** \$3
2. **Medium:** 2
3. **Hard:** 10
-

Unit rates and equivalent rates

g6_unit_rates_and_equivalent_rates · Grade 6 – Rates · gen=g6_unit_rates

Name: _____ Date: _____

Find the unit rate.

EASY

1. A student reads 24 pages in 4 minutes. Find the unit rate in pages per minute.

MEDIUM

2. A car travels 45 miles in 9 hours. Find the unit rate in miles per hour.

HARD

3. \$100 for 4 pounds of fruit. Find the unit price in dollars per pound.

Answer Key

1. Easy: 6
 2. Medium: 5
 3. Hard: \$25
-

Comparing ratios

g6_comparing_ratios · Grade 6 – Ratios · gen=g6_equivalent_ratios

Name: _____ Date: _____

Compare the ratios.

EASY

1. $\frac{2}{1} = \frac{4}{x}$

MEDIUM

2. Find the missing value: $11 : 8 = 22 : x$

HARD

3. Find the missing value: $12 : 29 = 36 : x$

Answer Key

1. Easy: 2
 2. Medium: 16
 3. Hard: 87
-

Equivalent ratios

g6_equivalent_ratios · Grade 6 – Ratios · gen=g6_equivalent_ratios

Name: _____ Date: _____

Find an equivalent ratio.

EASY

1. Find the missing value: $5 : 8 = 10$
 $: x$

MEDIUM

2. Find the missing value: $2 : 4 = 4 :$
 x

HARD

3. Find the missing value: $11 : 5 = 55 : x$

Answer Key

1. Easy: 16

2. Medium: 8

3. Hard: 25

Introduction to ratios

g6_introduction_to_ratios · Grade 6 – Ratios · gen=g6_introduction_to_ratios

Name: _____ Date: _____

Write the ratio.

EASY

1. Write the ratio $7 : 5$ as a fraction in simplest form.

MEDIUM

2. There are 6 blue books and 5 orange books. Write the ratio of blue to orange.

HARD

3. Write the ratio $22 : 2$ as a fraction in simplest form.

Answer Key

1. **Easy:** $\frac{7}{5}$
 2. **Medium:** 6 : 5
 3. **Hard:** 11
-

Part-part-whole ratios

g6_part_part_whole_ratios · Grade 6 – Ratios · gen=g6_introduction_to_ratios

Name: _____ Date: _____

Write the ratio.

EASY

1. There are 6 red apples and 8 yellow apples. Write the ratio of red to yellow.

MEDIUM

2. Write the ratio 4 : 8 as a fraction in simplest form.

HARD

3. Write the ratio 15 : 18 as a fraction in simplest form.
-

Answer Key

1. **Easy:** 6 : 8
 2. **Medium:** $\frac{1}{2}$
 3. **Hard:** $\frac{5}{6}$
-

Evaluating algebraic expressions

g6_evaluating_algebraic_expressions · Grade 6 – Variables and Algebraic Expressions · gen=g6_negative_number_operations

Name: _____ Date: _____

Evaluate.

EASY

1. $-9 - 5$

MEDIUM

2. $10 - 4$

HARD

3. $36 + -10$

Answer Key

1. **Easy:** -14

2. **Medium:** 6

3. **Hard:** 26

Writing algebraic expressions

g6_writing_algebraic_expressions · Grade 6 – Variables and Algebraic Expressions ·
gen=g6_writing_algebraic_expressions

Name: _____ Date: _____

Write the expression.

EASY

1. the product of 7 and a number

MEDIUM

2. a number divided by 9

HARD

3. a number increased by 9

Answer Key

1. **Easy:** $7x$

2. **Medium:** $\frac{x}{9}$

3. **Hard:** $x + 9$